

An Event-Dated Corpus of US Trademark Prosecution and a Two-Sided Measure of Vocabulary Position

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Abstract

All 13.99 million USPTO trademark case files since 1884 are re-parsed into 242 million dated prosecution events, so that every mark can be followed from filing through examination, registration, the sworn five-year proof of continued use, and the year-ten renewal, with counsel of record, filing basis and owner domicile read from the record, and owners linked by name to SEC reporting, Form D rounds and PatentsView assignees. Event dating separates the year-six use proof from the year-ten renewal, which terminal status codes conflate, and makes the proof a dated, product-level survival outcome available for every registered mark. On this corpus a two-sided measure of vocabulary position is built. Each goods/services description is reduced to themes by a topic model fitted once over the corpus, and its theme mix is compared, by Kullback–Leibler divergence, to its own Nice class in the five years before and the five years after filing. The average of the two divergences is atypicality; their difference is lead. The measure is validated on identical filings under three partitions of the vocabulary, two reference weightings, five window constructions, a second model seed, and the text duplication that pervades a boilerplate-heavy corpus. Lead is robust to all of them and predicts cancellation for non-use at the five-year proof (+2.3, +2.6 and +1.9 points under the three partitions; +1.4 with design controls, $t = 8.3$). Atypicality is not: its apparent effects belong to the partition or to a control. Two hazards for trademark-text research are documented. Scoring words rather than theme distributions measures description length ($r = -0.68$ with log distinct terms), and firm-level correlations built on word scoring dissolve or reverse under theme scoring on identical samples. Within firm, the measure is orthogonal to patenting. The corpus, the scores, the code, and an interactive per-class appendix are public.

Keywords: trademarks; administrative data; text as data; topic models; Kullback–Leibler divergence; innovation indicators.

JEL: C81, O34, O31, L10.

1 Introduction

A US trademark application carries a description of the goods or services the mark will cover. It is written to secure scope of protection, it is dated, it is public, and it is filed by the millions of firms that never patent. The literature that reads trademarks as innovation data has used the filing’s count, its class, its breadth and its renewal (Castaldi, 2020; Dinlersoz et al., 2018; Flikkema et al., 2019); the description itself has entered only piecemeal, as breadth counts or similarity scores. This paper builds two research assets on the description and on the prosecution record that surrounds it, states what each can and cannot support, and makes both public.

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The first asset is a corpus. The USPTO publishes every application since 1884 with its prosecution history, and terminal status codes are what the literature has read from it. Status codes conflate events that the maintenance schedule keeps apart: a registration cancelled for failing the year-six proof of continued use and one cancelled at the year-ten renewal share a code. Re-parsing all 13.99 million case files into 242 million dated events assigns each cancellation to the proof it failed by its date, and records for each filing the office actions, the responses and their latency, counsel of record, the statutory filing basis, and the owner’s domicile. The five-year proof of continued use thereby becomes a dated, sworn, product-level survival record available for every registered mark more than seven years old. Owners are linked by normalized name to SEC reporting, to Regulation D rounds, and to PatentsView patent assignees.

The second asset is a measure. Each description is reduced to themes by a topic model fitted once over the whole corpus, and its theme mix is compared, by Kullback–Leibler divergence, to what its own Nice class filed in the five years before and the five years after its date. The average of the two divergences is the filing’s *atypicality*, how unusual its language is for its industry; their difference is its *lead*, whether its language sat ahead of or behind where the industry’s language was moving. The two-sided reference is what separates *unusual* from *early*, and the two components predict different things.

The measure is validated by variation rather than by a single fit. The identical filings are rescored under three partitions of the vocabulary (fifty global themes, five hundred, fifty fitted per class), two reference weightings, five window constructions, a second model seed at fixed resolution, and with and without the 47% of registrations whose text is shared verbatim with another filing. Lead survives every variation and predicts failure at the five-year proof: +2.3, +2.6 and +1.9 points from the most lagging to the most leading fifth under the three partitions, +1.4 points with class-by-cohort fixed effects, drafting and filer controls and owner-clustered errors ($t = 8.3$). Atypicality does not: its protective effect at the proof falls from 4.4 points under global themes to 0.7 under per-class themes, because a global model reads a theme imported from another class as atypical, and its sign at the SEC-reporting margin depends on how description length is held.

Two hazards follow for any research that scores trademark text, and both are documented on identical samples. Scoring words rather than theme distributions measures description length: the divergence decomposes into a surprise term and a spread term, the spread term is bounded by the filing’s distinct-term count, and term-scored atypicality correlates with log distinct terms at -0.68 . Firm-level correlations built on word scoring, which are the kind the literature reports, dissolve or reverse under theme scoring on the same firms; the signal they carry is unsigned lexical specificity, not vocabulary position. A third caution concerns the sample rather than the score: applicants who refile an abandoned application rewrite toward their class’s typical language from both tails, so every registered description is in part a conformed one, and pooling granted with abandoned filings mixes the examiner’s selection with the market’s.

What the measure predicts about pioneering, about industry waves, and about which firms reach the capital markets is the subject of a companion paper; here those outcomes appear only as validation, and the paper makes no claim about mechanism. Section 2 describes the corpus, the steps of the record and the three outcomes it dates. Section 3 builds the measure and shows, on named filings, what themes resolve and what they miss. Section 4 settles nomenclature against the innovation literature’s existing terms. Section 5 tests the measure against registration, the five-year proof, patenting and SEC reporting. Section 6 reports what changes under every alternative construction tried. Section 7 states the practices that follow for trademark-text research, and Section 8 describes the public artifacts.

2 The record: from case files to dated events

2.1 Corpus and parse

The USPTO Trademark Annual Applications backfile (product TRTYRAP) is the complete record of US trademark filings from 1884 onward: 13.99 million case files, each carrying the goods/services description by class, the prosecution history, and the ownership and correspondence records. The prosecution history is a list of coded events with dates. This paper re-parses every case file into 242 million dated events, keyed by serial number, so that for each application the sequence of office actions, responses, allowances, statements of use, registration, Section 8 and Section 9 filings, cancellations and renewals is available with its date.¹ From the same files the parse reads counsel of record on each filing (and, for refilings, on each attempt separately), the statutory filing basis (use, intent to use, §44(e) home registration, §66(a) Madrid extension), the owner’s domicile, and the owner name, which is normalized — uppercased, stripped of punctuation and of two dozen legal suffixes — so that ACME CORP. and ACME CORPORATION are one owner. Graham et al. (2013) describe the administrative data; the event-dated parse adds the full prosecution timeline of each case.

Response latency in prosecution is a by-product of the parse and is retained as a health indicator: the median days from an office action to the applicant’s response, per filing.

2.2 The steps, the samples, and the conventions

The record’s steps, the outcome names, and the counting conventions are set out once, here, and every later section uses them.

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a sworn statement of use is filed — the proof that the mark has entered commerce — the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period — the *five-year proof of continued use*, the paper’s central outcome; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a proof is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific proof by its date rather than inferred from a status snapshot.²

Three named outcomes recur (Table 1). *Failing to register*: the application never completed, which happens by the applicant’s own inaction in roughly nine cases of ten (Section 5.1). *Failing the five-year proof of continued use*: a registration cancelled for non-use at registration age 4.0 to 8.5 years, which means the product the mark named left commerce within six years (Section 5.3). *Reaching the SEC record*: the owner appears in dated SEC events — a priced Form D round, financial reporting, a listing (Section 2.5).

¹Event codes are read against the USPTO’s own event-code dictionary. Terminal status codes are retained for cross-checking: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired. The year-six cancellation and the year-ten cancellation share status 710 and are separable only by event date, which is the reason the corpus is event-dated.

²Cross-checked against the USPTO status-code dictionary: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; five-year-proof outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in this section.

Stage	<i>n</i>	Share
Debut filings scored, 1995–2018 (owners’ first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the five-year proof of continued use (event-dated)		52.3%
Passed the five-year proof, cohorts 2002–2013, status known	1,129,724	
failed at the year-ten renewal		39.1%

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single Nice class and year so that no difference between industries or between years enters; regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in the axis, in percentage points.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner’s first, one per owner. A *cohort* is the set of registrations issued in a calendar year; where a split is instead by filing year, the text says so.

Two windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019: a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020, and the 1995 cut is about two years more conservative than the measured burn-in requires (Section 3.7). Five-year-proof cohorts run 2002–2018: cancellations post at about registration age six and a half, the record ends in April 2026, so 2018 is the last cohort whose window has essentially elapsed, and 2002 is the conventional lower cut for this corpus — a bound that matters, because carried back to the first scoreable cohorts the lead penalty was larger than it has ever been since, as the companion paper shows.

2.3 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations file differently and fare differently:

	Share of filings	Reached registration	Median terms (009 / 035)
Represented	73.3%	61.9%	28 / 26
Self-filed	26.7%	45.6%	21 / 17

Self-filers write shorter descriptions (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class’s usual language, and lean slightly

further ahead of it. Because who drafts moves both the language and the outcome at once, counsel of record is held in every regression in the paper, and the later sections report their estimates separately by stratum. The supplementary material draws the full flow from filing to registration split by drafter.

2.4 Office actions and refileing: applicants rewrite toward the class

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. Scoring each registration on the text it was first tried with and on the text it registered with gives the direction of the change. Both self-filers and represented owners rewrite toward the class’s typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (supplementary material). Lead barely moves once the reference windows’ shift with the later filing date is taken out.

Figure 1 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section 2.4 whose text changed. A pair is an abandoned application and the same owner’s later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

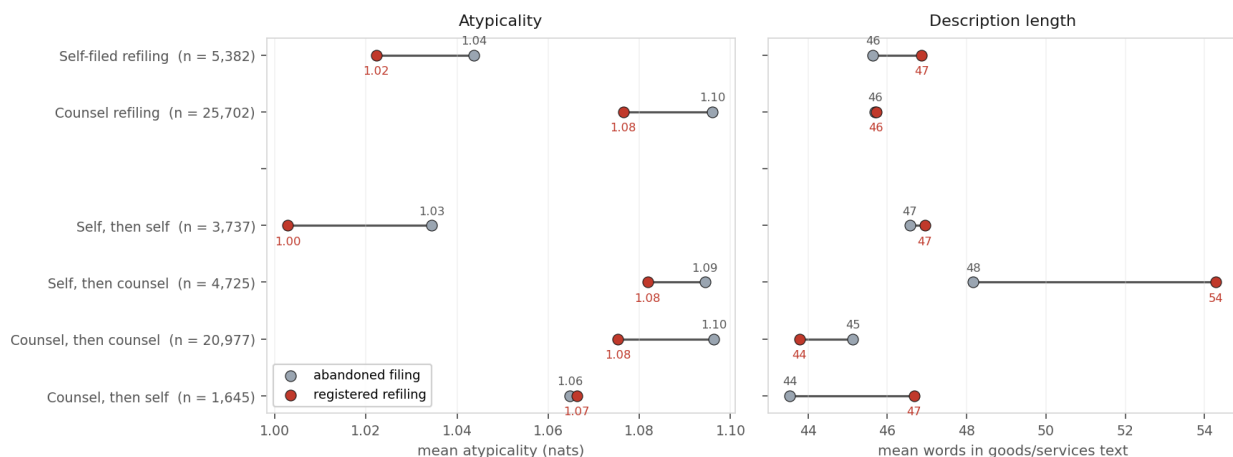


Figure 1: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

The means understate the convergence, because it runs in both directions. By fifth of the

original’s atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

2.5 Owner links: SEC reporting, Regulation D rounds, patents

Owners are matched to three external records by normalized exact name. *SEC reporting* is any appearance of the owner as a registrant in the Commission’s EDGAR filer index; roughly half the matched firms carry a ticker and the rest report for other reasons, and the two halves are distinguished by the Commission’s company-ticker file. *Regulation D* rounds are Form D notices of exempt offerings, electronic and therefore complete only from 16 March 2009, so any analysis of a funding ladder is built on debuts from 2009. *Patents* are PatentsView assignee records (PatentsView, 2024), matched on the same normalized name.

The SEC match runs over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22; a random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate. The linkage is nonetheless a name match, and that bounds what can be read from it: firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text. Section 5 returns to which results survive that bound.

3 The measure

3.1 Two readers, one living backwards

Imagine two readers of a filing’s goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry’s language moved that way.

Both windows are anchored on the filing’s own date, spanning the 1,826 days before it and the 1,826 days after; both exclude the filing date itself, so a filing never appears in the reference of another filing made the same day, and no two filings made on different days share a reference. Inside a window every day counts the same. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

3.2 What is measured

The quantity is information content in Shannon’s sense. Something that happens with probability p carries $\log(1/p)$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing’s own industry drew on each of them in the reference window. A filing’s score is the average

information content of its themes under its industry’s frequencies, weighted by how heavily it uses each, less the same average taken under the filing’s own frequencies.

The apparatus is that of Murdock et al. (2017), developed on Darwin’s reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. Their object was the life of ideas: how often an idea appeared, and how ideas stood in relation to one another, across one reading life or one assembly’s debate. The object here is the composition of commercial offerings, which components a product or service is described as having, and how that composition shifts across an industry over time. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author’s own reading, so a high value says something about a mind; here a filing is scored against other firms’ filings, and its author is usually counsel drafting for scope of protection. The same shift renames what the model finds. In those settings, and in the name of the method, latent Dirichlet allocation identifies *topics* — what a document is about. A goods/services description is not about anything in that sense; it lists components, and across millions of filers a cluster of terms that co-occur (*video, music, games, electronic, audio*) is a component many offerings share — a theme the entrants’ offerings evidence, not a subject. *Theme* is the word used throughout; *topic* appears only in the model’s name.

3.3 From a description to two numbers

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 36 are in the theme model’s vocabulary, among them *computerized, search, ordering, ordering service, retail, distribution, printed books* and *digital transmission*.

Grouping words into themes. Each filing is then re-described in a fixed, much smaller set of coordinates. Latent Dirichlet allocation is run once over a stratified sample of the whole corpus, all 45 classes together: it learns 50 clusters of words that tend to occur together and, for any filing, the proportions in which its words draw on them. Nothing supervises this; the clusters come from co-occurrence alone. This build — fifty global themes, per-filing five-year windows, flat weighting — is the production scoring, used for every estimate unless the text states another. The themes are shared across classes; what is specific to a class is the reference, the average theme proportions of that class’s own filings in the window. The Amazon filing draws on six themes at more than a twentieth of its weight: 0.41 on media and entertainment goods (*video, computer, electronic, music, games, audio*), 0.15 on computer and information services (*services, computer, software, information, data*), 0.13 on retail store services (*store, retail, store services, featuring*), 0.10 on printed matter (*paper, books, printed, cards, stationery*), 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods. The online appendix lists every theme with its leading words and its weight in each class.³

³https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html

Fifty is coarse, roughly one theme per industry. Section 5.3.3 refits the model at 500 themes and separately at 50 themes within each class and repeats the five-year-proof analysis under each; Section 6 reports the resolution sweep and a second fit at the same resolution with a different seed. Fifty is kept elsewhere because it is the coarsest resolution at which every result holds, and because its software-class lead contrast sits at the low end of the resolutions tried (only the 100-theme refit returns less).

Comparing against the two references. Write P for the filing’s distribution over themes, and Q^- and Q^+ for that distribution averaged over every same-class filing in the two windows, $[d_i - W, d_i]$ and $(d_i, d_i + W]$, with $W = 1,826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\text{KL}(P \parallel Q) = \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \text{KL}(P \parallel Q^-), \quad K^+ = \text{KL}(P \parallel Q^+),$$

so K^- is what the first reader feels and K^+ the second: the filing’s past-facing and future-facing surprise.

Taking the difference. Amazon’s filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A = \frac{1}{2} (K^- + K^+)}_{\text{atypicality}}, \quad \underbrace{L = K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$: against its own class, its atypicality is at the 47th percentile and its lead at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Table 3 maps the terms and their nearest occupants in the literature; the signed difference is written L throughout and its magnitude $|L|$.

What “early” means is visible in the themes themselves: the three themes that carry most of Amazon’s description — retail store services, media and entertainment goods, and the still-small online-software theme — all became more common in class 035 over the following five years, the last nearly doubling, while the two minor themes it touched receded. The filing is closer to the class’s future composition than to its past one; the supplementary material tabulates the six theme shares.

K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice. Taking their average and their difference — nothing more than a sum and a difference, with no weights estimated — isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated (+0.036 against each other, where K^- alone would give +0.114). The cost is that L is a small residual of two large quantities, about a sixth of either level’s spread, which Section 6 quantifies.

Why the words are not scored directly. The themes are an extra step, and the words themselves could be compared to the reference: build the filing’s distribution over the class vocabulary from the words it uses, do the same for every filing in the window, and take the divergence. The

paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. The divergence is an average surprise less a spread: for any two distributions,

$$\text{KL}(P \parallel Q) = \underbrace{-\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} - \underbrace{(-\sum_k P_k \log P_k)}_{\text{how spread out the filing is}},$$

and the spread of a filing over words is bounded by how many distinct words it has: a description using k terms cannot spread over more than k of them, so its spread cannot exceed $\log k$, while the reference is spread over about 900 terms. A five-word description is charged the whole difference in width, roughly five nats, before any of its content is considered. In 3,000 software-class filings of at least 150 words, scored against one fixed reference and rescored after truncation to a random handful of words, across a fiftyfold cut in length the surprise term moves by 0.016 nats and the spread term by 2.9; synthetic filings drawn at random from the reference itself track $\log(\text{distinct terms})$ to within a fiftieth of a nat (supplementary material). Term-scored atypicality correlates with the log of a filing’s distinct term count at -0.68 in the software class.

Themes break the coupling because the 50 coordinates are always occupied. Under term scoring the effective number of coordinates a filing spreads over rises seventeenfold from the shortest descriptions to the longest; under themes it varies by less than a third, and not monotonically: a three-word fragment gives the model little evidence, so its inferred proportions stay close to the prior and spread across many themes. The spread term no longer tracks length, and the score is dominated by the surprise term. Theme-scored atypicality correlates with the log term count at -0.22 in the software class and -0.10 in advertising and retail, against -0.68 and -0.62 under terms.

What the themes see, and what they do not. The score compares a filing’s theme proportions to the average proportions of its industry, so it reads which themes a filing draws on and how heavily. It does not, in any material way, read how a filing combines them. Taking each filing’s two leading themes and asking how often that pair occurs together relative to the two themes’ separate frequencies, on a 150,000-filing software-class sample: how rare the themes are individually explains 70% of the variance in atypicality, and the unusualness of the pairing adds under two points on top of it. A filing assembled from ordinary parts in a genuinely new configuration is therefore close to invisible to this measure.

Combinations can be scored in their own right, at two levels (construction detail in the supplementary material). A *pair lead* — whether the class moved toward the filing’s arrangements of themes — is penalised like lead in the software class (+2.3 points at the five-year proof, $t = 8.4$, correlating with lead at only -0.16), but it does not generalize: corpus-wide it is zero raw and +0.5pp net of lead and atypicality (owner-clustered $t = 4.2$), significantly positive in 13 of 43 classes and significantly negative in 3, including -2.1 in advertising and retail, so it is reported as a class-specific bound rather than a second construct. Word-level *recombination* — adjacent word pairs new to the class while both words are established there, novelty assembled from familiar components (Fleming, 2001; Uzzi et al., 2013) — is rewarded in the same class: the top fifth fails 2.8 points less with lead and atypicality held ($t = -10.4$), uncorrelated with pair lead ($r = -0.01$). The construct sees neither arrangement channel, and where they act they run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction.

3.4 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language from the USPTO’s own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors’ successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations issued 2002–2018 share a value with another filing in the same class and registration year. Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal: a leading filing was ahead of its industry’s language, which implies a trajectory without asserting the filing caused it.

3.5 Term scoring measures length; theme scoring does not

Figure 2 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. AMAZON.COM (1997) is at the 98th percentile of lead on terms and the 94th on themes, but only the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange. Apple’s IPOD (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Theme scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which theme scoring does not. Hence theme scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the theme basis has no coordinate for the language the examples turn on.

3.6 How three filings encode

Table 2 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

AMAZON.COM (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention (“computerized line,” “line search,” “search ordering”; class document frequencies 379, 134, 355). The filing’s text reads “on line” as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword “on” and the bigram “line search” survives stopword removal. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in

the theme vocabulary but, with no kombucha-like theme, the filing loads on a generic beverages theme at every T (Table 2).

ETHEREUM (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme; there, the eleven occurrences of *software* dominate the theme assignment.

Table 2: Top theme loading for three filings, by representation. Term scoring resolves the novel content of all three; theme scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccha- rose	blockchains, dis- tributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, devel- opment (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses theme scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Section 5.6).

3.7 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 3). The burn-in cut is set empirically: thin past references inflate past-facing surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of ~ 2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations (~ 0.15 – 0.17) are not burn-in, because mechanical contamination declines monotonically as the windows fill and cannot rebound. They reflect a genuine vocabulary shift in the corpus during the dot-com period. Per-class burn-in lengths estimated under a stability-band criterion track each class’s noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

4 Neighbours and nomenclature

4.1 Trademarks as innovation data

A trademark filing is a dated, costly, public disclosure made by the millions of firms that never patent, and a literature has grown around reading it as an innovation indicator (Castaldi, 2020;

Willeke et al., 2026). Trademark counts and breadth predict venture valuations (Block et al., 2014); filing timing tracks innovation processes (Seip et al., 2018); and trademarks mark the product-market side of innovation that patents miss (Flikkema et al., 2019, 2025). Census-linked work reads registration as a correlate of firm growth and survival (Dinlersoz et al., 2018, 2023). Nearly all of it counts filings, classes, or renewals; the descriptions’ text enters, if at all, as breadth or similarity. This paper reads the text itself, and reads it against time.

4.2 Text measures of novelty

Scoring a document by its divergence from a reference corpus before and after it comes from cultural analytics: Murdock et al. (2017) on Darwin’s reading notebooks, Barron et al. (2018) on the French Revolution’s parliamentary debates. In economics and finance, text-based measures locate firms in product space (Hoberg & Phillips, 2016), date technological innovation in patents over the long run (Kelly et al., 2021), and track the vintage of patent language (Kalyani, 2024). The construction here differs in its object — the composition of commercial offerings, not ideas or claims — and in its two-sided reference: every filing is scored against its own industry’s language both before and after its date, which is what lets *unusual* be separated from *early*. The separation matters because the two components predict opposite things.

4.3 Nomenclature

The innovation literature is not consistent about what quantities like these are called, and several of its established terms assert things this design cannot show, so the paper’s names are chosen against the nearest occupants rather than borrowed from them. Table 3 maps the five quantities; the entries below say, for each declined term, what adopting it would claim.

Table 3: The five quantities, their nearest occupants in the literature, and the terms used here. P is a filing’s theme distribution; the references Q^- and Q^+ aggregate the five years of same-class filings before and after it (Section 3). *Lagging* and *leading* describe the two signs of L . The level names appear only where the two-sided construction itself is at issue; the analysis runs on A and L .

Quantity	Nearest in the literature	Term used here
$K^- = \text{KL}(P Q^-)$	novelty (Barron et al., 2018; Murdock et al., 2017); originality (Hall et al., 2001)	past-facing surprise
$K^+ = \text{KL}(P Q^+)$	transience (Murdock et al., 2017); generality (Hall et al., 2001), which measures realized influence, not position	future-facing surprise
$A = \frac{1}{2}(K^- + K^+)$	atypicality (Uzzi et al., 2013), which scores internal pairings, not distance from the class	atypicality
$L = K^- - K^+$	resonance (Barron et al., 2018), which asserts a mechanism	lead (leading / lagging)
$ L $	no established occupant	lead magnitude

Novelty and *transience* are the two KL levels themselves; novelty is declined as a name for anything here because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is arithmetically what this paper calls lead, and is avoided because it names a mechanism

— the field resonating with an author’s contribution — that this design cannot observe: a filing can sit where its class was heading because it was imitated, because both responded to the same outside event, or by coincidence.

Radicalness is the closest neighbour among the innovation labels, since KL is a departure measure rather than a newness count. It is declined because it claims existing competences are devalued, which nothing here licenses — and because the label itself is unsettled: a bibliometric review of 4,407 articles finds *radical*, *breakthrough*, *discontinuous* and *disruptive* used with definitions inconsistent within labels and not distinctive across them (Verhoeven et al., 2025).

Atypicality in the bibliometric sense (Uzzi et al., 2013), applied to patents as unconventionality by Berkes & Gaetani (2021), scores the rarity of the pairwise combinations inside a document against the population’s co-occurrence rates. The atypicality here is a different object: the distance of one document’s composition from its class’s, not the rarity of its internal pairings; the two are close to independent on this corpus (Section 3). The word is kept because both mean unlike the population; which population, and along what, differ.

Originality and *generality* (Hall et al., 2001) are the patent literature’s standard backward- and forward-facing pair, and generality is the nearest established forward-facing quantity to K^+ . Both measure the dispersion of citations across technology classes, and generality requires the forward citations to have arrived, so it observes realized influence; K^+ observes the class’s movement whether or not the filing influenced it.

Distinctiveness has two prior occupants: trademark law’s registrability doctrine, and strategic management’s optimal distinctiveness (Zhao et al., 2017), which predicts an inverted-U that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not. *Innovation* is a property of the product, not of the sentence describing it. What *atypicality* says is narrower than any of these: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

5 Validation against the record’s outcomes

A measure is worth the outcomes it can be tested against. The record supplies four that are dated and high-stakes: whether an application reached registration, whether a registered mark’s owner proved continued use at year six, whether the owner patents, and whether the owner appears in SEC reporting. This section reports what each axis of the measure predicts at each, under the design that the corpus supports — class-by-year fixed effects, drafting and filer controls, owner-clustered errors — and states, for each result, whether it survives the alternative constructions of Section 6. The results are stated as validation of the measure. What they say about pioneering, and how they move across industries and decades, is the subject of the companion paper.

5.1 Registration rises with atypicality and is nearly flat in lead

Figure 4 gives the registration rate by decile of each axis, with deciles cut within Nice class and filing year so that no difference between industries or between years enters. On atypicality the rate rises from 56.0% in the least atypical decile to 59.1% in the ninth and turns down to 58.3% in the most atypical — a three-point climb and a one-point turn at the top. The turn is larger in some industries than the pooled line shows: software and electronics rises through the whole range (51.8% to 60.6%), clothing dips in the middle deciles, and both end higher than they start. On lead the range is 0.9 points with no consistent direction: filings written in language their class was moving toward register at nearly the same rate as filings written in language it was leaving.

The flatness of lead at this step has a mechanical reading worth stating: lead is computed from the five years of filings *after* the filing date, which had not happened when the examiner acted. No participant in prosecution could observe the quantity. Whatever small association appears must run through correlates that were visible at the time — the words themselves, the industry, the drafter. Atypicality’s past-facing half, by contrast, is exactly what an examiner steeped in the class’s usual language experiences, and it is the axis registration responds to.

Figure 5 shows the same outcome on the two surprise axes directly, for the two largest classes: each hexagonal cell is colored by the share of its filings that registered. The color gradient runs along the diagonal — from the low-surprise corner, where registration is lowest, up through the middle of the range — and barely across it. Distance from the diagonal is lead; position along it is atypicality. The plane says graphically what the deciles say numerically: the examination step reads how far a filing sits from its class’s usual language, not which direction it faces in time.

Two cautions on reading the level. First, registration is mostly a measure of the applicant’s own follow-through: of the applications that do not register, roughly nine in ten die by inaction — the applicant never files the statement of use, or stops responding during prosecution — and only 4.3% are removed adversarially (Table 1). Second, cut on the pooled distribution rather than within class and year, and restricted to debut filings, the atypicality curve is an inverse-U with a deeper conventional tail (46.6% at the bottom, about 58% in the middle, 53.4% at the top); that version carries class and cohort composition, and the supplementary material reports it with the reasons familiarity alone cannot produce the shape. The within-cell profile of Figure 4 is the cleaner statement, and on it the conventional tail, not the atypical one, is the penalized end. The shape is the one audience-conformity research predicts for evaluators (Hsu, 2006; Zhao et al., 2017; Zuckerman, 1999), here from an examiner and at census scale; Taeuscher & Rothe (2021) find settings where the optimum sits off-centre.

5.2 Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 7, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 6A).

The signed axis behaves differently. Completion on L falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. The signed axis still enters a linear specification strongly (harmonized estimates in the companion paper), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing’s language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of -0.06 pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading.

5.3 The five-year proof of continued use: leading marks fail it more often

This section is about one outcome: whether a registered mark’s owner proved, between the fifth and sixth year of the registration, that the product the mark names was still in commerce. The sample is every unique registration issued 2002–2018 that carries a score; failure is a dated cancellation for non-use at registration age 4.0–8.5 years (Section 2.2). Half of registered marks fail it. How the penalty moved across industries and decades, and what became of the firms, is the subject of the companion paper.

5.3.1 Why the analysis conditions on registration

The question is about companies whose products entered commerce: given that a product was brought to market, does the language of its first description predict whether it was still there five years on? Registration is the record’s marker of entry — a use-based application asserts the mark is in commerce, and an intent-to-use application registers only after a sworn statement of use — so the analysis begins there. An application that never registered left the record, nine times in ten, by the applicant’s own inaction, before any product met a customer. Including those applications would change the question into one about everyone who considered launching, including those who then did not, and since they are 2.3 million of the 6.0 million scored applications, their inclusion also buries the commercial signal in launch noise. Section 6.7 shows the pooled analysis for a reader skeptical of the conditioning: the registration step’s shape is stamped onto every later outcome, and the gradients found below are attenuated and in places reversed.

Conditioning would miscount if abandonment were usually a pause rather than an ending — if owners reworked refused applications and refiled, the same product would be counted once as a failure and again as a success. It is not: among 2.28 million abandoned applications, 3.6% are refiled within six years and 1.5% lead to a registration (Section 2.4). Crediting every successful refile as an eventual registration moves the registration profile by less than a tenth of a point. The refiled cases are 1.1% of this section’s sample, they fail the five-year proof at nearly the sample rate (51.7% against 52.8%), and the headline contrast below is +2.25pp with them and without them.

5.3.2 Leading marks fail the five-year proof more often

Marks written in language their industry subsequently moved toward fail the five-year proof more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, registrations in the most leading fifth of lead fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 4).

Among 4.12 million registration class-records 2002–2018, dated cancellation at the five-year proof rises from 50.0% at the bottom lead quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 9). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with lead quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;⁴ and standard errors clustered on 1.27 million

⁴A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof

normalized owners (Table 4).⁵

Table 4: Failure at the five-year proof of continued use and lead: design-based estimates. LPM of event-dated failure on within-class×registration-year lead quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5–Q1 (pp)	SE	<i>n</i>
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with *n* in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of two dozen legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of outcomes at the five-year proof is 0.38 pooled across classes (0.42 in the single-class checks where it was first measured), so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the five-year proof at 69.6% against 46.8% for represented owners, and Section 2.3 showed they write shorter, more class-typical, slightly more leading descriptions, so without the control part of any lead gradient would be a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose failure rates are high for their own reasons.⁶ About forty percent of the raw pooled contrast is composition: +1.4pp design-based against +2.5pp raw. The penalty is also not constant over the sample: within counsel-represented US-domiciled owners the earliest third of cohorts carries none

until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

⁵Tied scores are common (17.1% of registrations share a lead value within their cell); tie-breaking is deterministic and immaterial, moving the contrast by at most 0.002pp against 2.287 (supplementary material).

⁶The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

of it (-0.25pp , SE 0.33) — leading marks registered 2002–2007 were no less likely to survive than lagging ones — while the later thirds carry $+2.00$ and $+2.65\text{pp}$. The companion paper takes that variation as its subject.

The per-class extremes of the raw screen are in the supplementary material and the online appendix.

The penalty is a rate, not a timing, and not disengagement. The proof is close to a fixed appointment (96.1% of cancellations post at registration age 6.5–7.0), so a mark failing outside the 4.0–8.5 window would count as a survivor; on the 3.15 million registrations whose ninth anniversary is inside the record, where failure is settled and no window is needed, the contrast is $+2.71\text{pp}$ against the window’s $+2.65$, and conditional on failing, failing late carries a contrast of -0.01pp ($t = -0.2$). Nor is the proof measuring an owner who lost interest: passage is flat across office-action response-speed quintiles (46.1% to 48.1%, medians seven days to 183) and the penalty is present in both speed halves (-3.48 and -2.36pp on passing, against -2.94 pooled). The supplementary material carries the detail, including why 2019 is the last observable cohort.

5.3.3 The penalty under three theme scorings

Every estimate above scored filings on fifty themes fitted once over the whole corpus, with each class supplying its own reference. That is one of three ways to partition the vocabulary, and the penalty does not depend on the choice. The identical registrations were re-scored under *500 global themes* (the same construction at ten times the resolution, so a theme is closer to a product category than to an industry) and under *50 themes fitted per class* (a separate model on each class’s own filings, so nothing is shared across classes). In all three, surprise is measured against the filing’s own class on per-filing windows of 1,826 days each side.

The lead penalty is a property of the filing, not of the partition (Figure 10, left): $+2.29\text{pp}$ under the production scoring, $+2.61\text{pp}$ at 500 themes, and $+1.94\text{pp}$ under per-class themes, each with a standard error of 0.09, and positive under all three at once in 28 of the 44 classes with at least 3,000 sampled registrations. The scorings agree only moderately filing by filing — lead correlates at 0.64 between the two global resolutions and at 0.52 between the global and the per-class fifty — so a third to a half of the variance in any one scoring’s lead is particular to that scoring, and the penalty survives the disagreement. (Table 5 tabulates the contrasts; the standard errors here are two-sample binomial and unclustered, and outcomes correlate within owner at 0.38, so single-class t -statistics overstate precision by roughly a factor of two.)

Table 5: Failure contrasts at the five-year proof under three theme scorings, on the identical registrations. Top-minus-bottom quintile, quintiles cut within Nice class and registration year, no controls; $n = 3,105,048$, base failure 52.3%.

Scoring	Lead Q5–Q1 (pp)	SE	Atypicality Q5–Q1 (pp)	SE
Global, 50 themes	+2.29	0.09	−4.39	0.09
Global, 500 themes	+2.61	0.09	−4.29	0.09
Per class, 50 themes	+1.94	0.09	−0.72	0.09

Atypicality behaves differently (Figure 10, right). Under the two global scorings the most atypical fifth of registrations fails the five-year proof 4.3–4.4 points less often than the least atypical; under per-class themes the gap is 0.7 points. The two global scorings’ atypicality correlates at 0.79; global and per-class atypicality correlate at 0.39. A global model places every class in one set of coordinates, so a filing that imports a theme the class rarely uses and other classes use often

registers as atypical for its class. A model fitted on the class alone re-partitions the class’s own vocabulary, and the imported theme becomes a local theme like any other. What the global scoring calls atypical is, in large part, drawing on a theme from elsewhere.

That population can be measured directly, and it separates two kinds of new. Under the 500-theme model, for each theme and class, the first sustained year is the first year by which the theme has been the dominant theme of at least one hundred filings in that class; the corpus-wide first sustained year is the earliest of those across classes.⁷ Each registration in the sample is assigned its dominant theme and classified at its filing date as carrying a theme new to the corpus, a theme new to its class (established elsewhere, arriving here), or an established theme. Of 4.12 million registration class-records, 3.97 million carry an established theme, 151,012 a theme new to their class, and 779 a theme new to the corpus — the last too few to support a claim, because almost every theme a 500-theme model can resolve had been established in some class before 2002.

Registrations carrying a theme new to their class fail the five-year proof at 48.3% against 51.6% for established themes, and 0.43 points less often within class and registration year (SE 0.13). Holding the filing’s own lead and atypicality as well reverses the sign: net of them, the new-to-class registrations fail 1.06 points more often. The two statements describe one population. A theme arrives in a class in filings that are atypical for that class, and atypical filings survive; that is the -0.43 . Among filings of equal atypicality, the ones carrying the arriving theme fail more; that is the $+1.06$. The atypicality protection of the global scorings is therefore mostly the protection of extension: language established elsewhere and extended into a new class survives. The penalty attaches to being early with language, not to being far from the class’s centre.

Nor is the lead penalty an atypicality effect in disguise. The two could be confounded mechanically — lead is the difference of two nonnegative levels whose average is atypicality, so its spread grows with atypicality and its magnitude correlates with it at $+0.29$ — but the signed value does not (-0.05), and cutting lead quintiles within fifths of atypicality leaves the penalty intact in every fifth: $+2.1$, $+3.1$, $+3.3$, $+1.7$ and $+2.5$ pp from the least atypical fifth to the most (each SE under 0.3).

5.3.4 The risk persists at the year-ten renewal

Survivors of the five-year proof face a combined declaration and renewal at year ten, and on the design-based estimates the lead penalty does not attenuate there: -0.49 pp per standard deviation on passing the five-year proof, -0.63 pp on renewing at ten (harmonized estimates in the companion paper, both stated on survival). Pooled across classes the renewal profile is a V rather than a slope — renewal is highest just below the centre of the lead axis and lowest at both tails, the lagging tail lowest of all (45.5% against 50.6% at the most leading; fit detail in the supplementary material). Most of the difference between leading and lagging resolves at the five-year proof; what remains at renewal is a penalty on both extremes, the lagging one the larger.

5.3.5 What the five-year proof shows

Among products that entered commerce, being described in language the industry was moving toward predicts being gone within six years: $+1.4$ points on a 52.3% base with the full design, $+2.3$ raw, in three of every four classes, under every partition of the vocabulary, robust to timing, to drafting, and to who files. Atypical language predicts the opposite, but that protection belongs mostly to filings extending a theme established elsewhere into a new class; net of atypicality, even

⁷Sustained years and theme shares are computed on a 25% systematic sample of all filings (every fourth serial), scaled up, over filing years 1986–2024.

those arriving-theme filings fail more. The market’s step and the examiner’s step read the text differently for their own reasons: the examiner reads distance from the class’s usual language at a moment when lead does not yet exist to be read (Section 5.1); the five-year proof is recorded after the industry’s language has moved, and it is where being early shows up as a cost.

5.4 The construct is not patenting

Within the firms that do both, trademark vocabulary and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), each axis is regressed on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm, estimated on 94,181 firm-years across 17,506 firms under theme scoring (93,551 and 17,415 under term scoring) — 134,075 matched firm-years before dropping the firm-years that contribute no within-firm variation. The outcome is standardized, so β is the shift in standard deviations of the axis per unit of log-patents; one unit is roughly a tripling of the patent count, and a firm going from none to twenty moves about three units.

Text counted as	Axis	β (sd)	t	Between-firm r
Themes	atypicality	+0.012	+3.0	—
Themes	lead	−0.058	−9.4	−0.020
Terms	atypicality	−0.055	−11.5	—
Terms	lead	+0.012	+2.0	—

The grid is close to a mirror: whichever axis carries the small signal under one way of counting the text carries the opposite sign under the other. The theme-scored *lead* coefficient (−0.058) and the term-scored *atypicality* coefficient (−0.055) are nearly the same number attached to different quantities. Only those two cells are established; the mirror’s other diagonal is within noise, and the orthogonality conclusion rests on the magnitude bound — every cell under 0.06 standard deviations — rather than on any cell’s sign. Nothing here reaches a tenth of a standard deviation.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly (Seip et al., 2018), and no timing offset from $t - 2$ to $t + 1$ isolates a direction (−0.054 to −0.089 sd under theme scoring, +0.012 to +0.024 under terms); among owners who both patent and raise a priced round, neither event systematically precedes the other (median lag zero years).

The second objection is heterogeneity: a pooled near-zero could hide a positive slope where the two rights are complements offsetting a negative one where trademarks work alone. On a coarse, judgement-based sorting of classes into complements and substitutes — assigned in advance from the class definitions, nothing estimated — the slopes are −0.000 among complements, −0.026 among substitutes, interaction −0.008 ($t = -0.49$), and the complement classes do not agree in sign with one another (cell detail in the supplementary material). Neither retention supports the complementarity story, with the caveats that a coarse sorting can miss a finer-grained one and that conditioning on a successful assignee match pushes any interaction toward zero.

5.5 SEC reporting: the lead profile survives the design; the atypicality gradient is composition

The outcome here is the firm-level one that the representation hazard of Section 5.6 turns on: whether an owner ever appears in the Securities and Exchange Commission’s financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. Figure 11 draws the raw rate against each axis in twenty bins. In

signed lead the profile is a U whose tails both sit above the base rate and whose lagging tail is the higher; in lead magnitude and in atypicality the raw rate rises.

The raw atypicality gradient does not survive the design-based estimates (Table 6). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by -0.003 pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition: firms that go on to report cluster in particular classes, years, and multi-class filings, and class $\times$ year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (-0.047 pp per standard deviation, $t = -7.8$) while the unsigned component enters positively ($+0.067$ pp per standard deviation, $t = 7.7$). The U in the raw bins is exactly this pair: lagging filings show the magnitude premium with no lead penalty, leading filings show it net of that penalty, and the middle bins show neither. Once the levels are controlled, the recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5.

Table 6: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class $\times$ filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class $\times$ year. Base rate 0.481%.

	Coefficient (pp)	SE
<i>Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ (Q1 omitted)</i>		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	-0.003	0.018
<i>Signed lead quintiles, FE + log length + KL levels</i>		
Q3	-0.133	0.020
Q4	-0.187	0.021
Q5	-0.158	0.024
<i>Decomposition, FE + log length</i>		
lead magnitude $ L $ (pp per sd)	+0.067	0.009
signed lead L (pp per sd)	-0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 4.

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. The lead coefficient is unchanged under alternative linkages; the atypicality coefficient is not, so only the lead result is robust to the name-match design.

The U is also not atypicality working through lead’s construction (the mechanical coupling stated in Section 5.3.3): on this sample lead magnitude and atypicality correlate at +0.25, signed lead at only −0.10, and cutting lead fifths within fifths of atypicality keeps the structure: the most lagging fifth out-reports the most leading in four of the five atypicality strata (by 0.08 to 0.39pp on sub-1% bases), and the ordering reverses only in the most atypical fifth, where the most leading debuts report most. The lagging tail’s premium is not a ceiling artifact; the leading tail’s partial recovery in the raw U is where the atypicality composition lives, which is what the level controls in Table 6 absorb.

“SEC reporting” is not “went public”: splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U in lead (Section 6), so the result is not an artifact of counting non-operating registrants as successes.

5.6 Firm-level outcomes depend on the representation

The mark-level results above hold under theme and term scoring alike. The firm-level results do not, and the divergence is the second hazard this paper documents. Under term scoring, leading filings appear to belong to better firms; under theme scoring, on identical samples, every signed version of that finding dissolves or reverses. What follows reports the term-scored results as they would be published, then what happens to each under distribution scoring, and then why.

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term- L from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term- L predicts gross margin at +2.2pp per standard deviation ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with past-facing surprise entering positively and future-facing negatively; and firm-year term- L predicts excess stock returns of +1.5 to +5.0pp per standard deviation at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed theme- L is flat to declining, and the margin coefficient reverses: −1.3pp per standard deviation ($t = -1.8$) at $T = 50$, −2.2pp ($t = -3.1$) at $T = 200$, and −0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the past-/future-facing decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms’ counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The theme model’s vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, theme scoring would erase a real signal rather than an artifact. Computing each filing’s invisible-vocabulary share (63.8% of class-009 term *types* but only ~9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed L gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term- L mechanically. Theme count is a partition dial, not a coverage dial — a crypto/blockchain theme exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

6 Robustness: what changes under other choices

Each result in Section 5 is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

6.1 Representation and weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary’s terms. The *weighting* is how days inside the reference window count: every day equal, or decaying by half every two years. Every reference is per-filing, anchored on the filing’s own date (Section 3). Four builds exist:

Representation	Weighting	sd(lead)	sd(atypicality)
Themes	flat	0.102	0.698
Themes	half-life 2y	0.082	0.698
Terms	flat	0.155	1.101
Terms	half-life 2y	0.130	1.102

All four are compared on the 6.01 million filings every one of them scored. The first row is the production scoring.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Resolution, 50 vs 200 themes ⁸	0.636	0.828
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the 45-degree rotation of Section 3 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution

⁸The resolution row is computed on the five-year-proof sample rather than the 6.01 million-filing common sample of the other rows.

is a real test, which is why the term-scored replication in Appendix 6.10 and the sweep below are the ones the claims lean on.

6.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry; whether the findings depend on that coarseness is testable by refitting at higher T . Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Proof contrast, lead	t	Proof contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: it does not move monotonically in T , so no single resolution can be quoted as the software class’s effect, and the figure this paper reports at $T = 50$ sits at the low end of the range rather than the high one. A fifth fit at $T = 1000$ exists and is excluded: its first run shipped undertrained models through an indentation bug, and the refit is not persisted. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper’s estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

6.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may be unstable, with two runs settling in different local optima, in which case the sweep measures fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical sample, the five-year-proof lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit’s lead score

carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; correcting for it would enlarge the estimates, not shrink them. And the spread of magnitudes across the resolution table above, which does not move monotonically in T , should be read as a band about a third of a point wide from fitting noise alone rather than as a property of any particular T .

The substantive claim does not depend on any of this. The five-year-proof lead contrast on the 2002–2018 registrations each build scores runs +2.29, +1.95, +3.56 and +2.92pp across the four builds in the order tabulated above, and the atypicality contrast -4.39 , -4.43 , -3.63 and -3.73 pp. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

6.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million gate registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom gate contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is -4.39 and -4.43 pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

6.5 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the five-year-proof quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

6.6 Mixing the two windows

“Leading” is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. Scoring every class

on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluating all five variants on identical samples — the gate columns on the 3.10 million registrations 2002–2018, quintiles within class×registration-year; the registration columns on the scored filings 1995–2018, quintiles within class×filing-year:

Variant	Past, future (years)	First-gate failure		Registration	
		Q5–Q1 (pp)	classes > 0	Q5–Q1 (pp)	Q3 – tails
Foresight	3, 7	+1.65 (0.09)	31 of 45	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 45	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 45	+0.01	+0.33

The symmetric five-year row is the paper’s FE-only estimate in Table 4, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the use requirement punishes distance from the class’s established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. The examiner’s response to lead, on this scoring, is close to nil.

6.7 Without conditioning on grant

Figure 12 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ~ 2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in L because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%); the caption carries the quintile figures. Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth -2.11 pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure 13.

6.8 Toggled filters, windows and parameters

Each analysis in this paper and in the companion paper was rerun under its natural parameter toggles, on the identical data. Nothing reverses a headline claim; two auxiliaries move.

Variant	Baseline $\rightarrow$ variant	Reading
Gate window 3.5–9.0 / 4.5–8.0	lead +2.29 $\rightarrow$ +2.29 / +2.28pp	window immaterial
Registration cohorts 2002–09	+2.29 $\rightarrow$ +0.98pp	the era swing
Registration cohorts 2010–18	+2.29 $\rightarrow$ +3.11pp	penalty concentrated late
Atypicality, both cohort halves	−4.39 $\rightarrow$ −4.55 / −4.28pp	stable
Internet pattern, narrow core	goods web gap 0.0 $\rightarrow$ +2.0pp	divergence holds
Convergence arrival 1% / 4%	early $\rightarrow$ late +4.4 $\rightarrow$ +1.6	+5.9 $\rightarrow$ +1.8 / +3.8 $\rightarrow$ +2.3
Surge doubling 1.5 $\times$ / 3 $\times$	in-surge net +1.3pp	+1.2 / +0.9pp
Wave entry window 4y / 8y	order flat	flat under both
Recombination floor 2 / 10	net −2.8pp	−1.2 / −3.6pp
Pair-lead presence 0.05	net +2.3pp	+3.1pp
Combination measures, class 035	009 values	recomb −3.8; pair-lead −1.4
Ladder, counsel / self debuts	reporting lead −0.19pp	−0.26 / −0.03pp
Ladder, 2009–13 / 2014–18	reporting atyp. +0.11pp	0.00 / +0.19pp
Excluding duplicate-text filings	lead +2.29; atyp. −4.39pp	+3.05; −5.68pp
Excluding Manual-scale texts only	same	+2.64; −5.05pp

Three of these change what a reader should carry. The lead penalty at the gate belongs to the 2010–2018 cohorts; the 2002–2009 half carries a third of it, which is the era swing the companion paper documents, seen from the other side. The ladder’s gradients belong to counsel-represented debuts — among self-filers the reporting gradient is a null — and its atypicality gradient to the 2014–2018 half. And the pair-lead penalty, +2.3pp in the software class under every parameter tried there, is −2.1pp in advertising and retail on the corpus-wide construction, and pooled across 43 classes is zero raw and +0.5pp net of lead and atypicality, so it is reported as a bound for one class rather than a general fact. All artifacts are under `paper/v3/_eval/`.

6.9 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ ($n = 1,109,643$). The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level’s spread. Only 1.2% of the two levels’ combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels; under term scoring the dominant such thing is document length, and Section 3 quantifies the coupling under both representations. The consequence for the lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. Appendix 3.5 shows this as a length surface over the two axes. Two qualifications. Theme scoring returns a finite value for about 65% of filings, so the two columns are not drawn from the same set; re-estimating the term-scored correlations on exactly the theme-scored subset barely moves them, which puts the contrast down to the representation rather than the sample. And “flat” means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A . The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Section 5.6). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT “most innovative” lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$),

+0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with L on the same panel (Pearson +0.010).

6.10 Term scoring as a cross-check

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4pp survival across quintiles under theme scoring and -5.7pp under term scoring on the identical sample (Figure 13). Under term scoring, gate survival *rises* in the raw KL level ($+6.5\text{pp}$ past-facing, $+7.6\text{pp}$ future-facing): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 5.5, at the mark level.

6.11 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure 14): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	ΔAIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex -1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

The winning form is a free-vertex $a + be^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents between 0.5 and 1.2 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4} , which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a V, linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything is multiplying.

Vertices are in standard deviations of the axis across filings.

The two rows for SEC reporting are the outcome Section 5.5 and Table 6 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the five-year proof are weak, and those two rows are better read as “no worse than linear” than as curvature.

The non-monotone profiles on signed lead decompose into parts of different symmetry (Figure 15). Each ventile’s deviation of the listing rate from the base is split, by sequential counterfactual prediction, into the part atypicality composition alone predicts, the part exact-duplicate text adds, a linear lead tilt, and a residual. The symmetric U is carried about half by atypicality composition — the spread of L rises with A , so both tails of the lead axis hold atypical filings and the middle holds typical ones — and about half by a residual that is itself symmetric in $|L|$:

the same conditional lead magnitude effect Table 6 estimates at +0.067pp per standard deviation. Duplicate-text composition adds almost nothing once atypicality is held, and the linear tilt is an order of magnitude smaller than either symmetric part (-0.07 pp per nat). The fitted vertices above follow from this arithmetic: a symmetric profile plus a negative tilt has its minimum shifted right of zero, and a survival hump its maximum shifted left, which is where the fits put them ($+0.24$ to $+0.41$, and -0.4 , standard deviations).

Excluding text-duplicate filings strengthens rather than weakens every headline gradient. Of the five-year-proof sample, 47% of registrations share their exact normalized description with at least one other filing; dropping them raises the lead contrast from $+2.29$ to $+3.05$ pp and the atypicality contrast from -4.39 to -5.68 pp, and dropping only the Manual-scale texts (clusters of 100 or more identical filings) gives $+2.64$ and -5.05 pp. Identical texts carry identical scores and cluster at the typical end of the axis, so they dilute both gradients; neither is created by them.

6.12 Listing against reporting

“Appears in SEC records” mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission’s own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at $+0.41$ and $+0.24$ standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.⁹

6.13 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.38 for gate survival pooled and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of the companion paper — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t -statistics overstate precision by roughly a factor of two, which the text of those sections now carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

⁹Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore “currently listed” rather than “ever listed,” which will misclassify older successes downward. And the base rates here are computed on the appendix’s own panel, which conditions on the debut filing carrying a score and so is smaller than the body’s sample.

7 Practices for trademark-text research

Three practices follow for research that reads trademark text. First, score distributions, not words. Word-level divergence is dominated by description length — on this corpus the correlation with log distinct terms reaches -0.68 — and firm-level performance correlations built on it dissolve or reverse under distribution scoring on identical samples. Any term-level novelty measure should be accompanied by the theme-level rescoring that costs one model fit. Second, treat the registered description as a conformed document. Applicants who refile rewrite toward their class’s typical language from both tails, whoever drafts the second attempt, so samples of registered marks understate the atypicality applicants first attempted, and estimates on registered text are estimates on prosecution-filtered language. Studies using description similarity or breadth inherit this compression. Third, the two components of surprise are not interchangeable and should not be pooled. Lead is robust to every partition, weighting, and window tried here; atypicality’s apparent effects can be artifacts of the partition (its protective effect at the five-year proof belongs to theme importation, visible only to global models) or of a control (its listing effect depends on how description length is held). A study that reports “novelty” from trademark text without separating the temporal from the cross-sectional component may attribute to one what belongs to the other.

For users of the classification itself, one caution: internet-bearing filings in goods classes survive at rates closer to the technology classes than to their own, so Nice classes used as industry controls do not partition the internet economy. And for the maintenance data, one asset: the five-year proof of continued use is a dated, sworn, product-level survival record covering every registered mark — an outcome the indicator literature has largely left unused, observable for any cohort more than seven years old at no cost beyond parsing the prosecution history.

8 Access

Everything described here is public at <https://github.com/ericssilver/tm-vocabulary>. The repository holds the corpus build (the parser from TRTYRAP case files to dated events, and the derived event, extras, and dictionary tables), the theme model and its vocabulary, the per-filing scores under every construction reported in Section 6, the owner-link tables, every analysis script, and every generated figure and table in this paper. The pipeline rebuilds from the public USPTO, SEC and PatentsView sources.

An interactive online appendix at <https://aporia.institute/tm-vocabulary/online-appendix/> lists every theme with its leading words and its weight in each Nice class, gives per-class breakouts of every outcome, and includes a viewer of each class’s registrations on the lead/atypicality plane. Researchers who want the five-year proof as an outcome without the text measure can take the event-dated cancellation table directly; those who want the measure on a different corpus can refit the theme model with the scripts provided.

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Declaration of competing interest

The author declares no competing interests. The author’s current employment is unrelated to this work, and the views expressed are the author’s alone.

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Data availability

The corpus build, all analysis code, every generated artifact, and an interactive per-class online appendix are public at <https://github.com/ericsilver/tm-vocabulary>. The underlying USPTO, SEC, and PatentsView sources are public.

Declaration of generative AI in the writing process

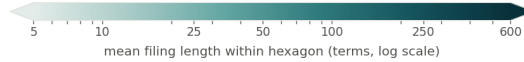
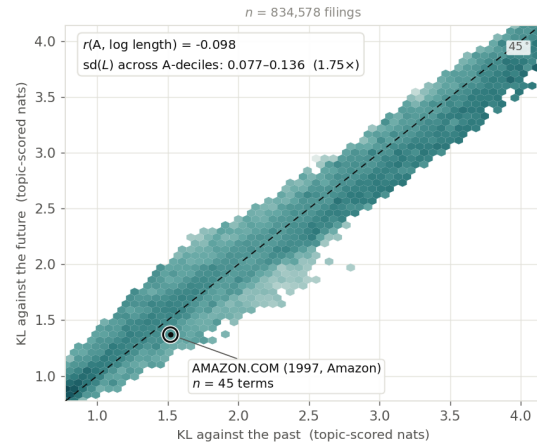
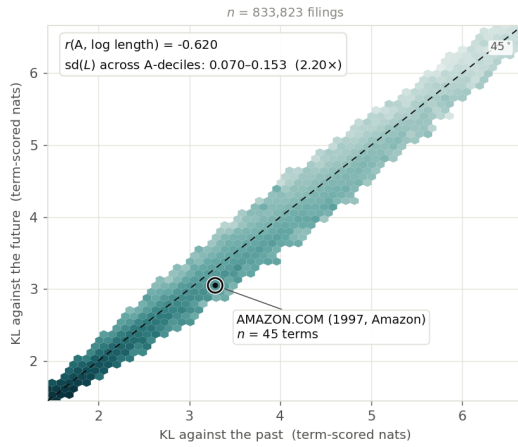
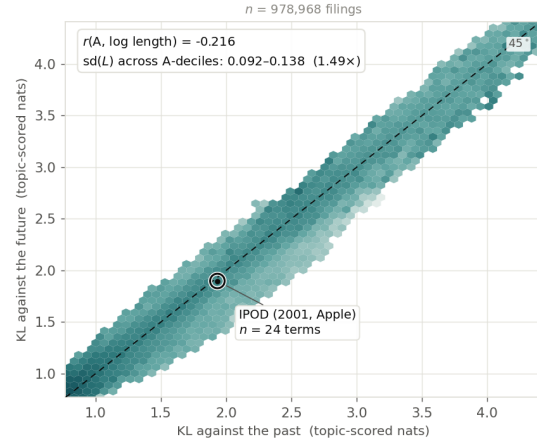
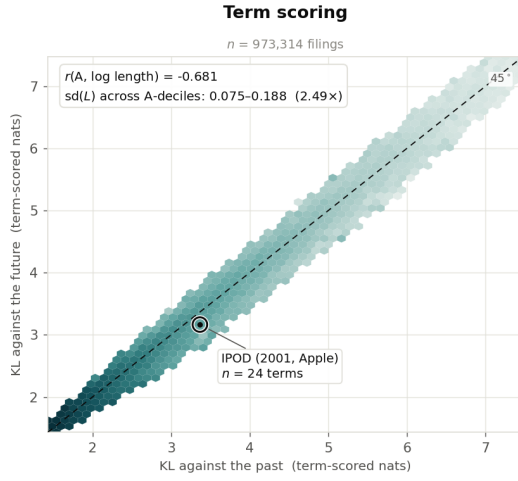
This paper was produced in a sustained collaboration with Claude (Anthropic), which drafted and revised text and analysis code under the author’s direction. Every analysis decision, every number against its artifact, and all final text were reviewed by the author, who takes full responsibility for the content.

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Hexagons holding fewer than 25 filings are left blank; the retained hexagons still cover over 99.4% of filings in every panel.
 The colour scale is shared by all four panels. The axis ranges are not: each panel is boxed on its own 0.5th–99.5th percentiles, and term-scored and topic-scored KL do not share a scale — the comparison invited across columns is of shape and colour gradient, not of raw magnitude.

Figure 2: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows. The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and theme-scored KL do not share a scale, axis ranges differ across columns because term- and theme-scored KL are not on a common scale, and only shape and colour gradient are comparable. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.

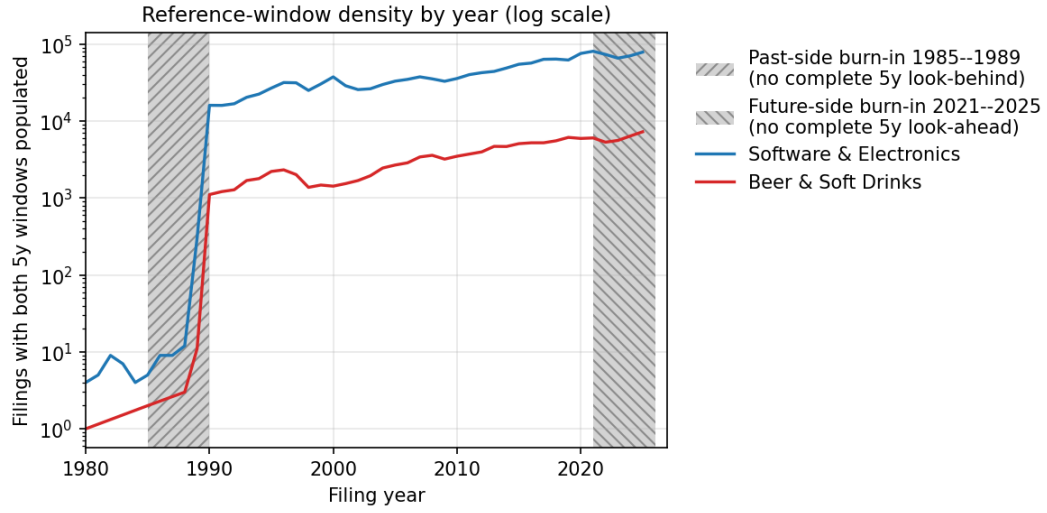


Figure 3: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The dotted line marks the 1995 analysis cut. The windows decide the score at the filing date; the gate outcome is read separately, at registration age six to seven.

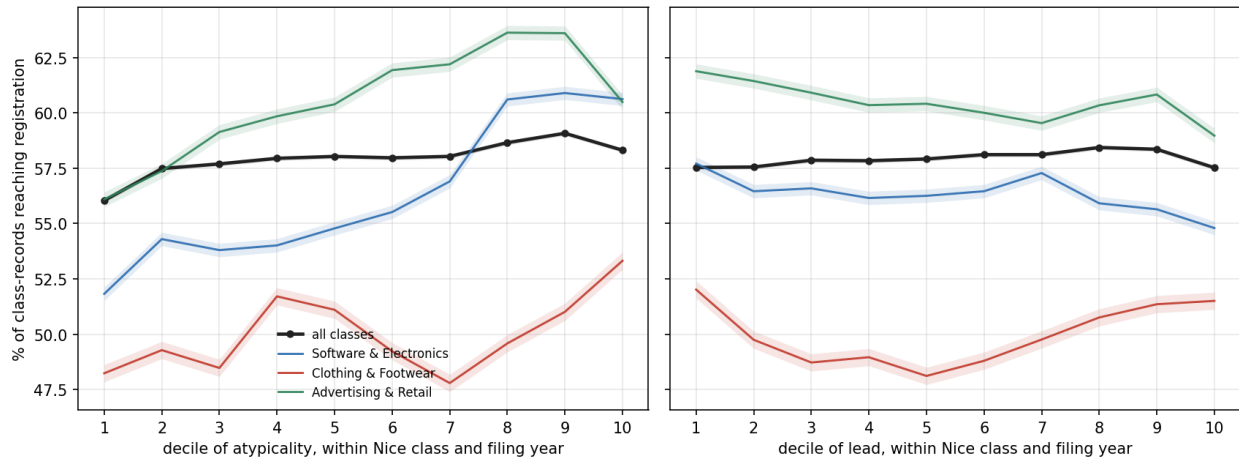


Figure 4: Share of class-records reaching registration, filings 1995–2018, by decile of atypicality (left) and of lead (right), deciles cut within Nice class and filing year; $n = 9,024,532$ scored class-records. The black line pools all 45 classes; the colored lines are three named classes. The grey band is a 95% binomial interval on the pooled line.

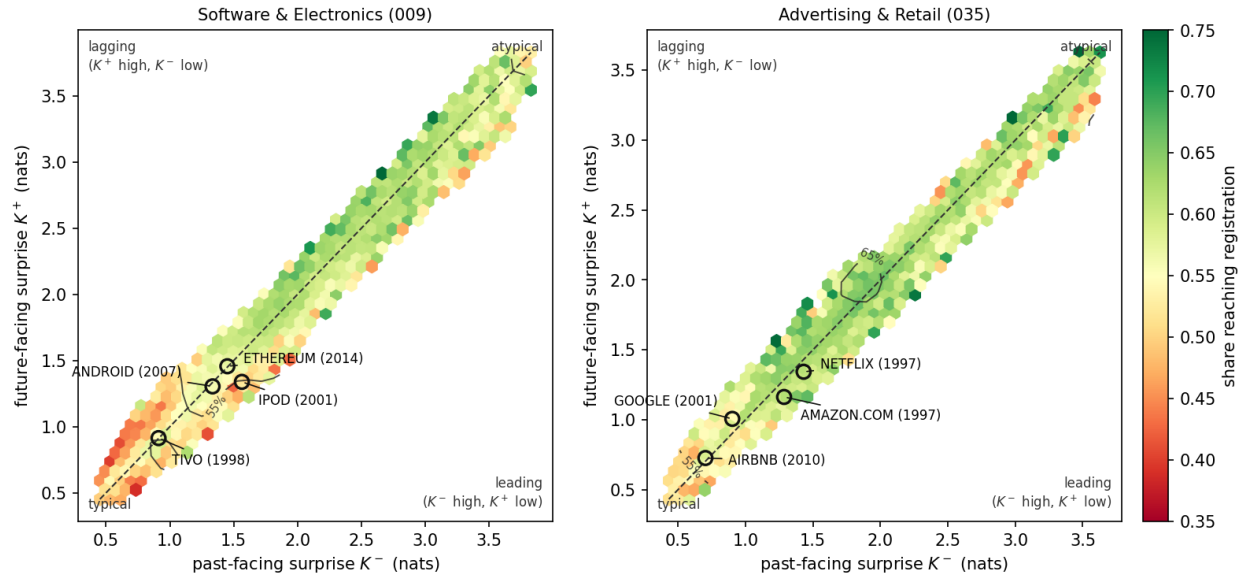


Figure 5: Share reaching registration across the surprise plane, filings 1995–2018 in the two largest classes. Each hexagon holds at least 50 class-records and is colored by the share of them that registered; the dashed line is $K^- = K^+$, on which lead is zero. Cells below the line are leading, cells above lagging.

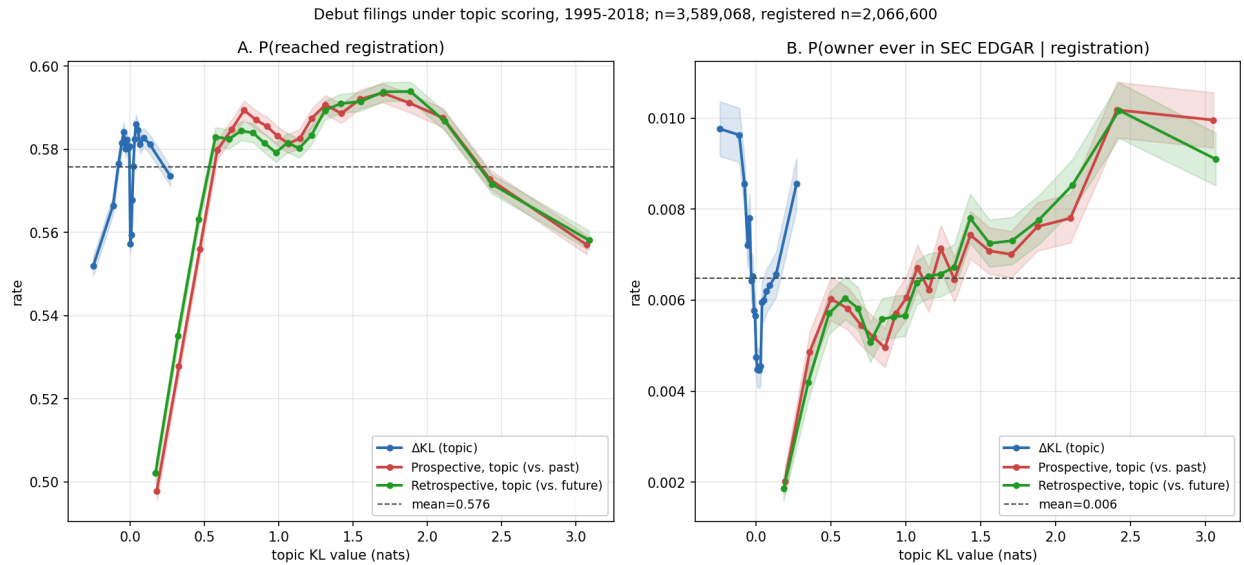


Figure 6: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals, quantiles cut on the pooled distribution. $n = 3,589,068$; registered subset $n = 2,066,600$. *A*: $P(\text{reached registration})$. *B*: $P(\text{owner ever in SEC EDGAR, the Commission's public filing system} \mid \text{registration})$.

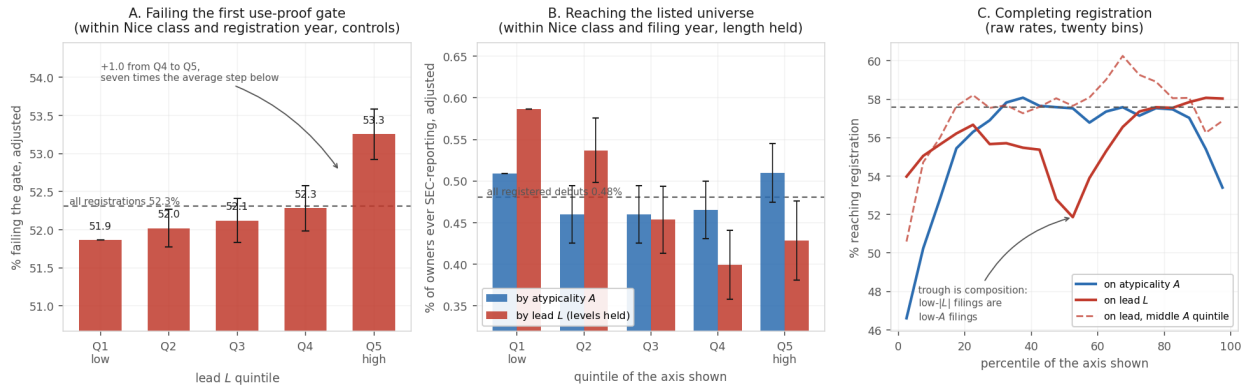


Figure 7: Quintile profiles for the three outcomes. *A*: the share of registrations failing the five-year proof of continued use, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 4; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. *B*: the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held. *C*: raw registration rates for debut filings in twenty bins, cut on the pooled distribution — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

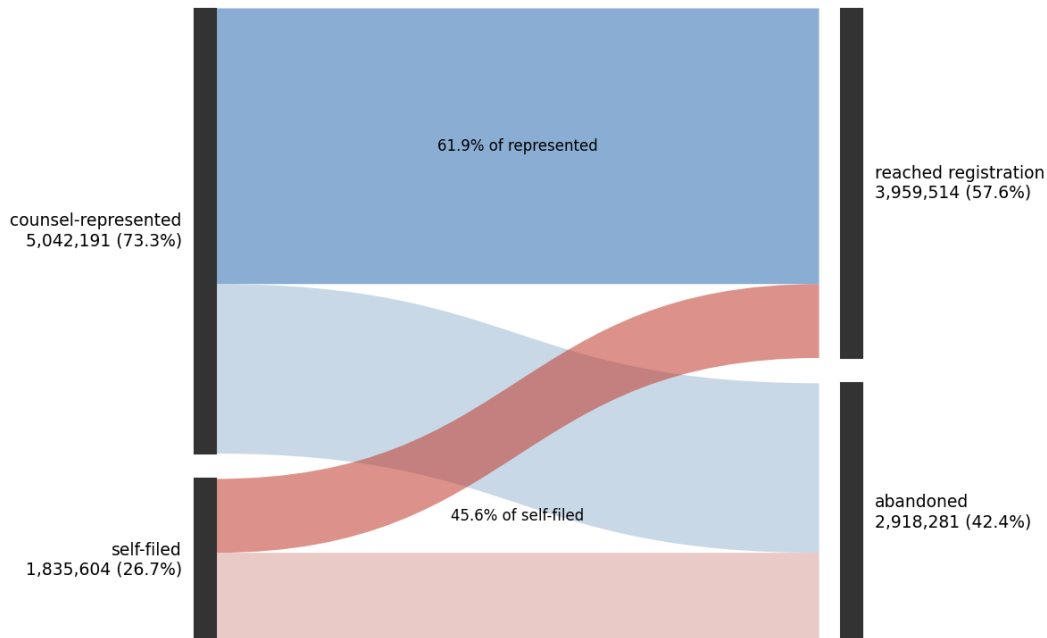


Figure 8: The registration step as a flow, unique serials filed 1995–2018 ($n = 6,877,795$), split by counsel of record on the filing. Represented applications reach registration at 61.9%, self-filed at 45.6%.

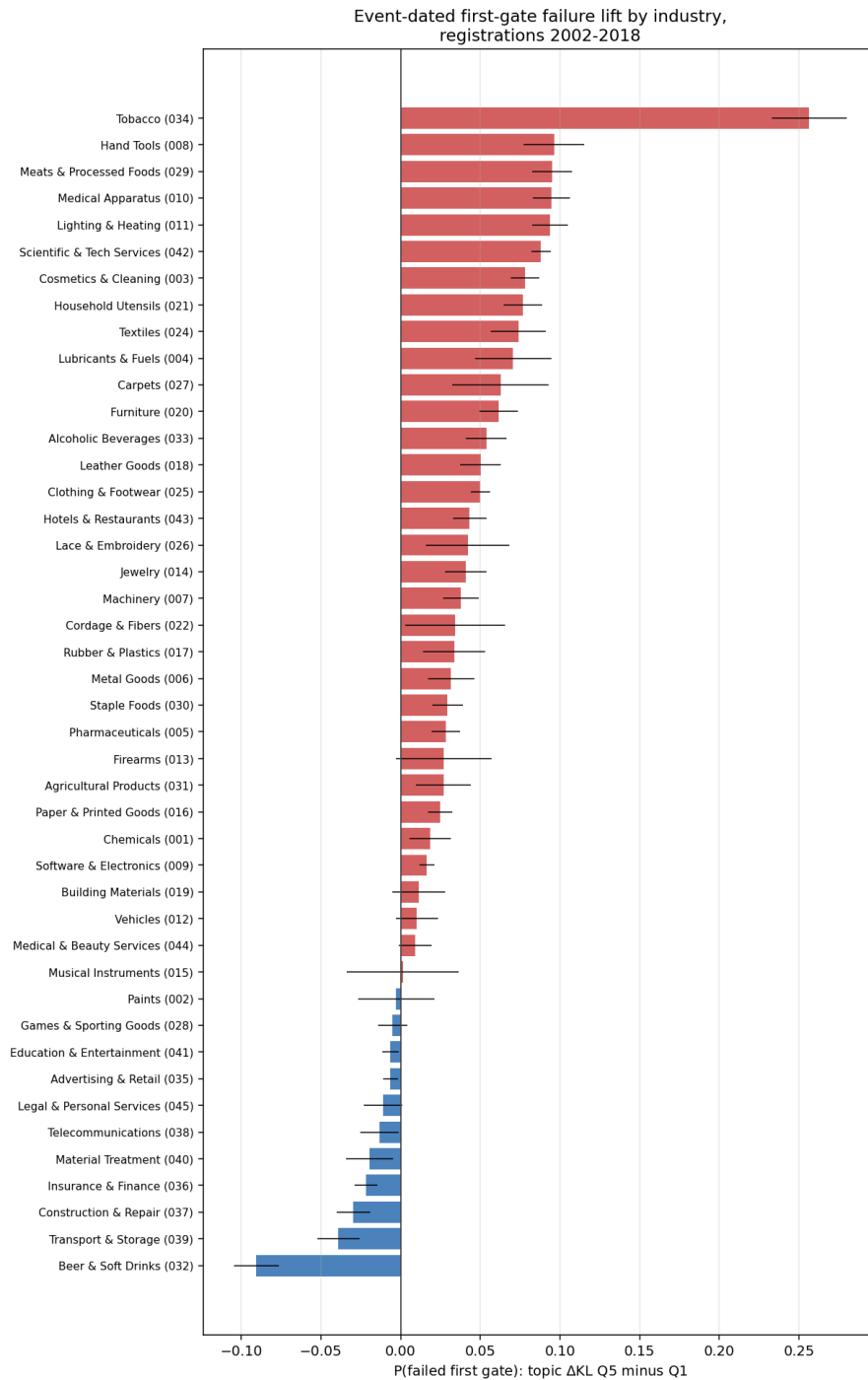


Figure 9: Failure lift at the five-year proof of continued use (lead top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts; classes need 5,000 scored registrations to enter. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.38) and so narrower than clustered intervals; read Table 4 for precision. Per-class breakouts are in the online appendix.

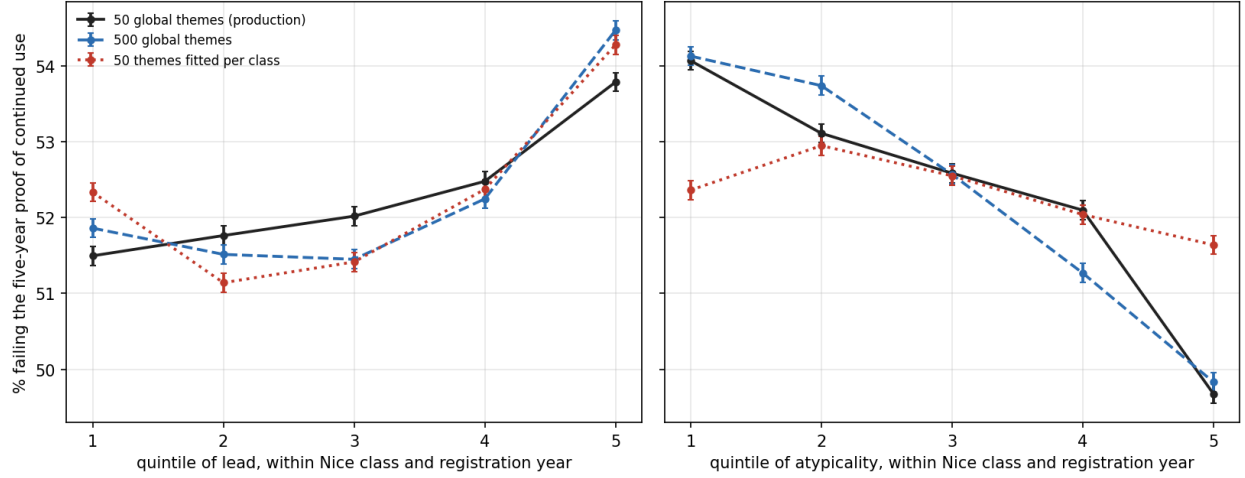


Figure 10: Failure at the five-year proof of continued use by quintile of lead (left) and of atypicality (right) under the three theme scorings, on the identical 3.10 million registrations 2002–2018; quintiles cut within Nice class and registration year, no controls. Whiskers are 95% binomial intervals per quintile, uncorrected for owner clustering. The lead gradient survives every scoring; the atypicality gradient does not survive per-class themes.

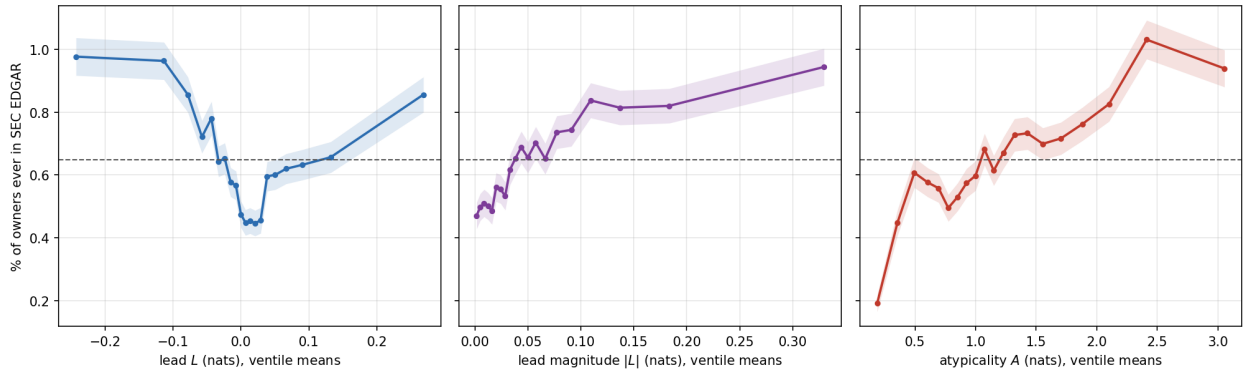


Figure 11: Share of registered debut owners (first registered filing 1995–2018, $n = 2,066,600$) ever appearing in SEC EDGAR, in twenty equal-count bins of debut lead L , lead magnitude $|L|$, and atypicality A , cut on the pooled distribution. Bands are 95% binomial intervals; the dashed line is the 0.65% base rate. These are raw rates carrying class, year and length composition; Table 6 gives the design-based estimates, under which the atypicality gradient is absorbed and the signed-lead U remains.

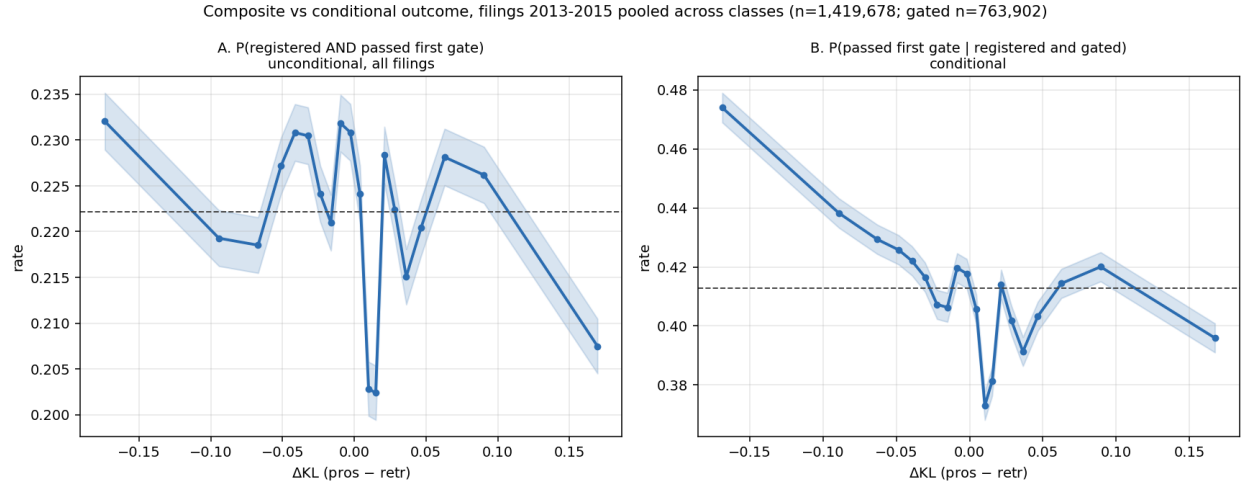


Figure 12: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows ($n = 1,419,678$; gated subset $n = 763,902$). *A*: the unconditional composite $P(\text{registered and passed the five-year proof})$ over all filings. *B*: $P(\text{passed} \mid \text{registered and gated})$ on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

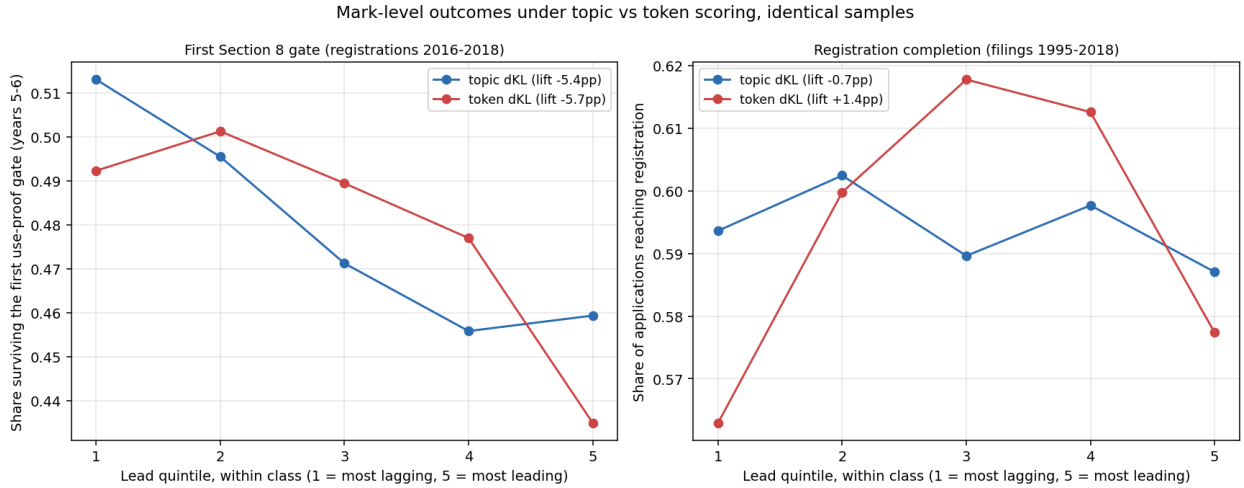


Figure 13: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left*: the share of registrations 2016–2018 that survived the five-year proof of continued use at years five to six, read from terminal status codes, where the two representations agree closely. *Right*: registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth -0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and the registration analysis reports only the unsigned version.

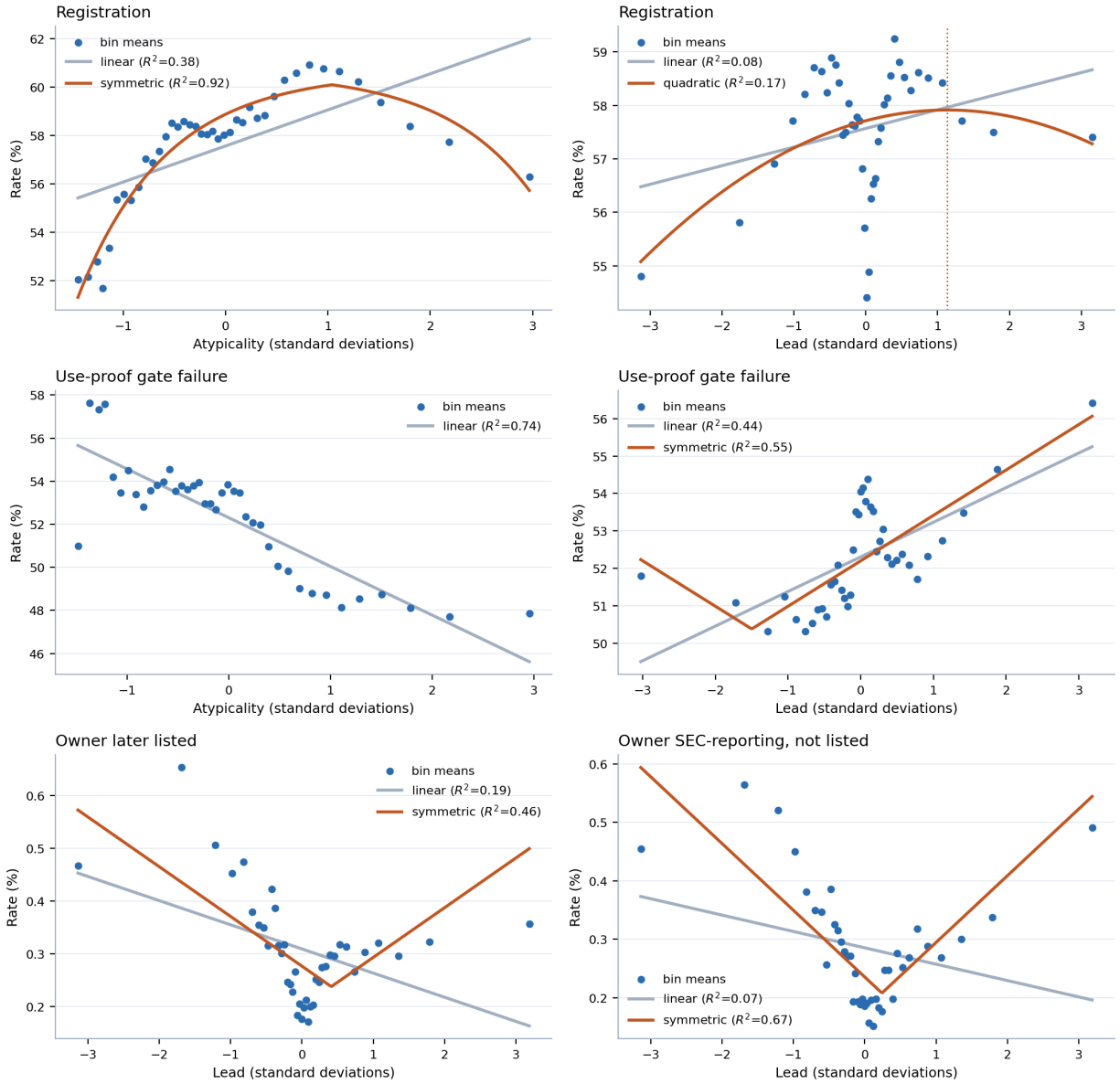


Figure 14: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

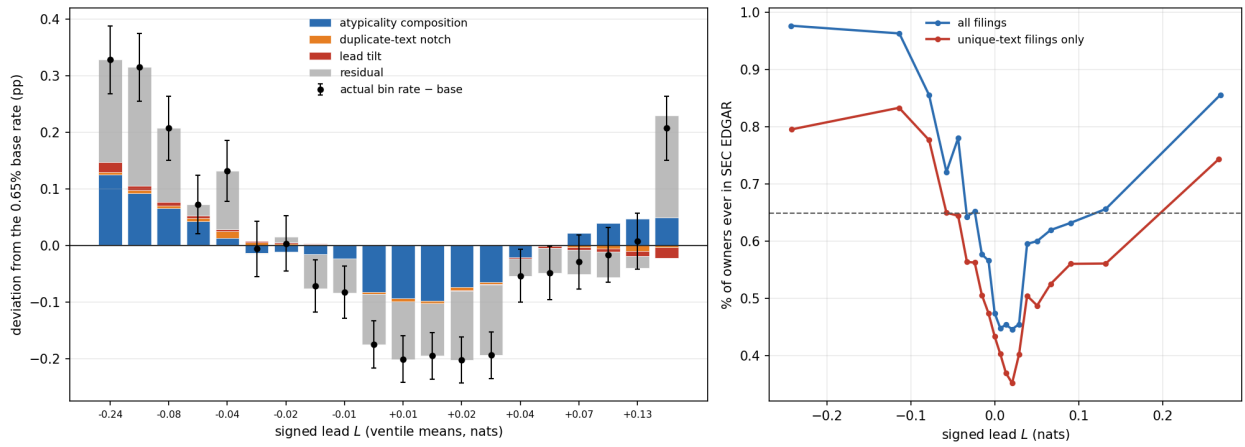


Figure 15: The signed-lead listing profile taken apart. *Left*: each ventile’s deviation of $P(\text{owner ever in SEC EDGAR})$ from the 0.65% base, on 2.07 million registered debuts 1995–2018, split into the part predicted by atypicality composition alone (50 quantile bins of A), the part added by an exact-duplicate-text indicator, a linear tilt in L fitted to the remainder, and the residual; bars stack from zero, dots are the actual deviations with 95% intervals. *Right*: the same profile on all filings and on filings whose normalized description text is unique in their class; the central dip survives, and deepens, without duplicated text.